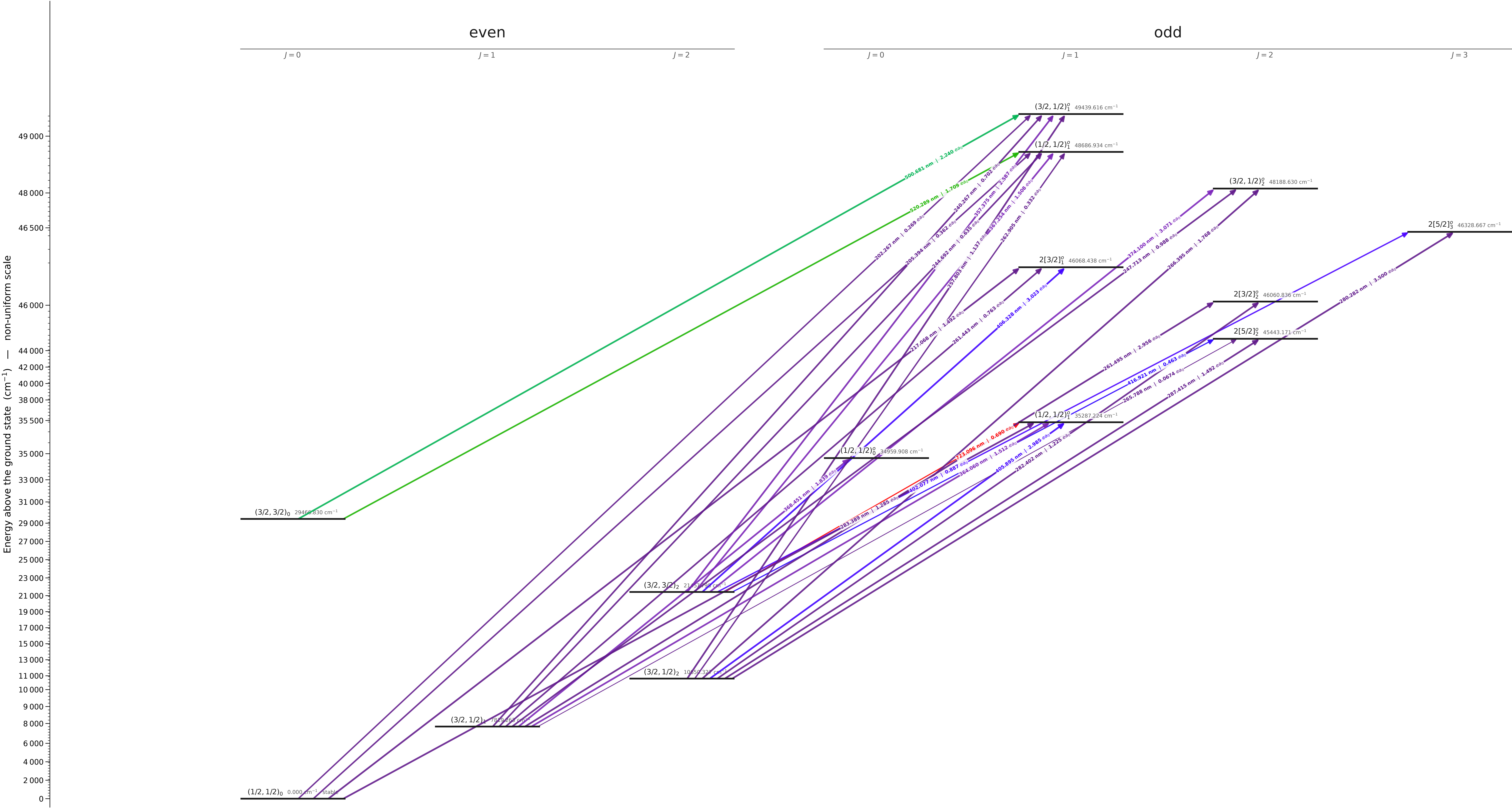
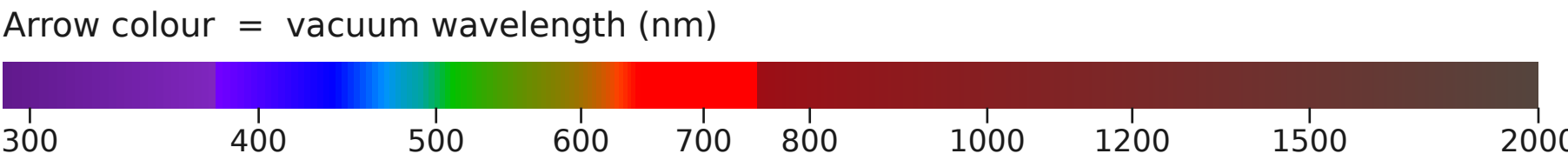


Lead

Pb (Pb I). Strongest classified transitions below 2 μm (NIST): 14 levels, 28 lines.

Pb ⁺ ionisation limit 59819.56 cm⁻¹ = 7.41668 eV (167.169 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|(J||e\mathbf{r}||J')|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e $\mathbf{r}$ J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
(1/2, 1/2) ₀ - 2[3/2] ₂ ^o	217.0684	217.0003	1381.0970	1.492	1.470e+08	-	-	-
(3/2, 1/2) ₂ - (3/2, 1/2) ₁ ^o	257.8031	257.7259	1162.8736	1.137	5.100e+07	-	-	-
(3/2, 1/2) ₁ - 2[3/2] ₂ ^o	261.4955	261.4175	1146.4535	2.956	1.980e+08	-	-	-
(3/2, 1/2) ₂ - (3/2, 1/2) ₂ ^o	266.3946	266.3154	1125.3700	1.768	6.700e+07	-	-	-
(3/2, 1/2) ₂ - 2[5/2] ₂ ^o	280.2821	280.1995	1069.6097	3.500	1.610e+08	-	-	-
(3/2, 1/2) ₂ - 2[3/2] ₂ ^o	282.4020	282.3189	1061.5804	1.225	2.700e+07	-	-	-
(1/2, 1/2) ₀ - (1/2, 1/2) ₁ ^o	283.3887	283.3053	1057.8844	1.285	4.900e+07	-	-	-
(3/2, 1/2) ₂ - 2[5/2] ₂ ^o	287.4154	287.3311	1043.0632	1.492	3.800e+07	-	-	-
(3/2, 3/2) ₂ - (3/2, 1/2) ₁ ^o	357.3749	357.2729	838.8738	2.587	9.900e+07	-	-	-
(3/2, 1/2) ₁ - (1/2, 1/2) ₁ ^o	364.0605	363.9568	823.4688	1.512	3.200e+07	-	-	-
(3/2, 3/2) ₂ - (1/2, 1/2) ₁ ^o	367.2537	367.1491	816.3090	1.508	3.100e+07	-	-	-
(3/2, 1/2) ₁ - (1/2, 1/2) ₀ ^o	368.4511	368.3462	813.6561	1.839	1.370e+08	-	-	-
(3/2, 3/2) ₂ - (3/2, 1/2) ₂ ^o	374.0998	373.9935	801.3702	3.071	7.300e+07	-	-	-
(3/2, 1/2) ₂ - (1/2, 1/2) ₁ ^o	405.8953	405.7807	738.5956	2.985	9.000e+07	-	-	-
(3/2, 3/2) ₂ - 2[3/2] ₁ ^o	406.3283	406.2136	737.8084	3.023	9.200e+07	-	-	-
(3/2, 3/2) ₀ - (3/2, 1/2) ₁ ^o	500.6813	500.5416	598.7691	2.240	2.700e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).